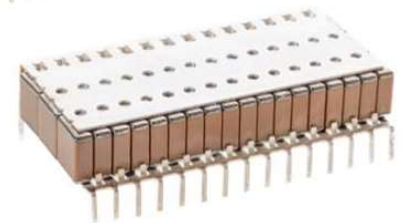
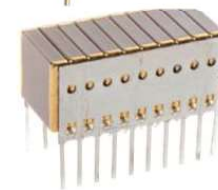
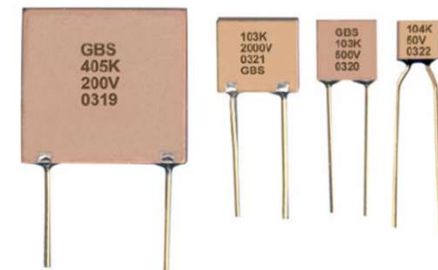
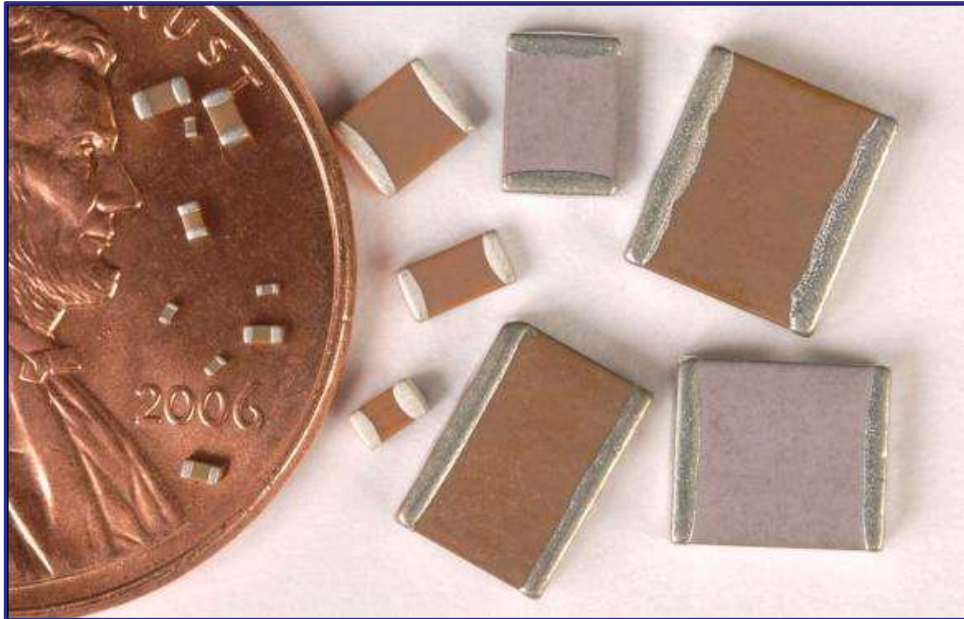


Electronic Components
KEMET
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**Ceramic Capacitors
(MLCCs)**

Design and Characteristics

Ceramic Chip Capacitors



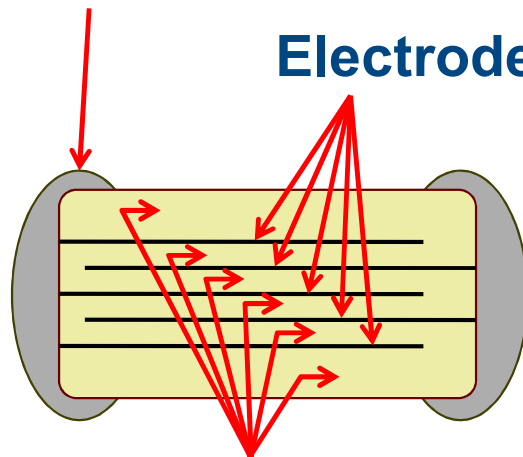


Design

Ceramic Capacitor Structure

Termination

Electrodes



Ceramic

C = Design Capacitance

K = Dielectric Constant

A = Overlap Area

d = Ceramic Thickness

n = Number of Electrodes



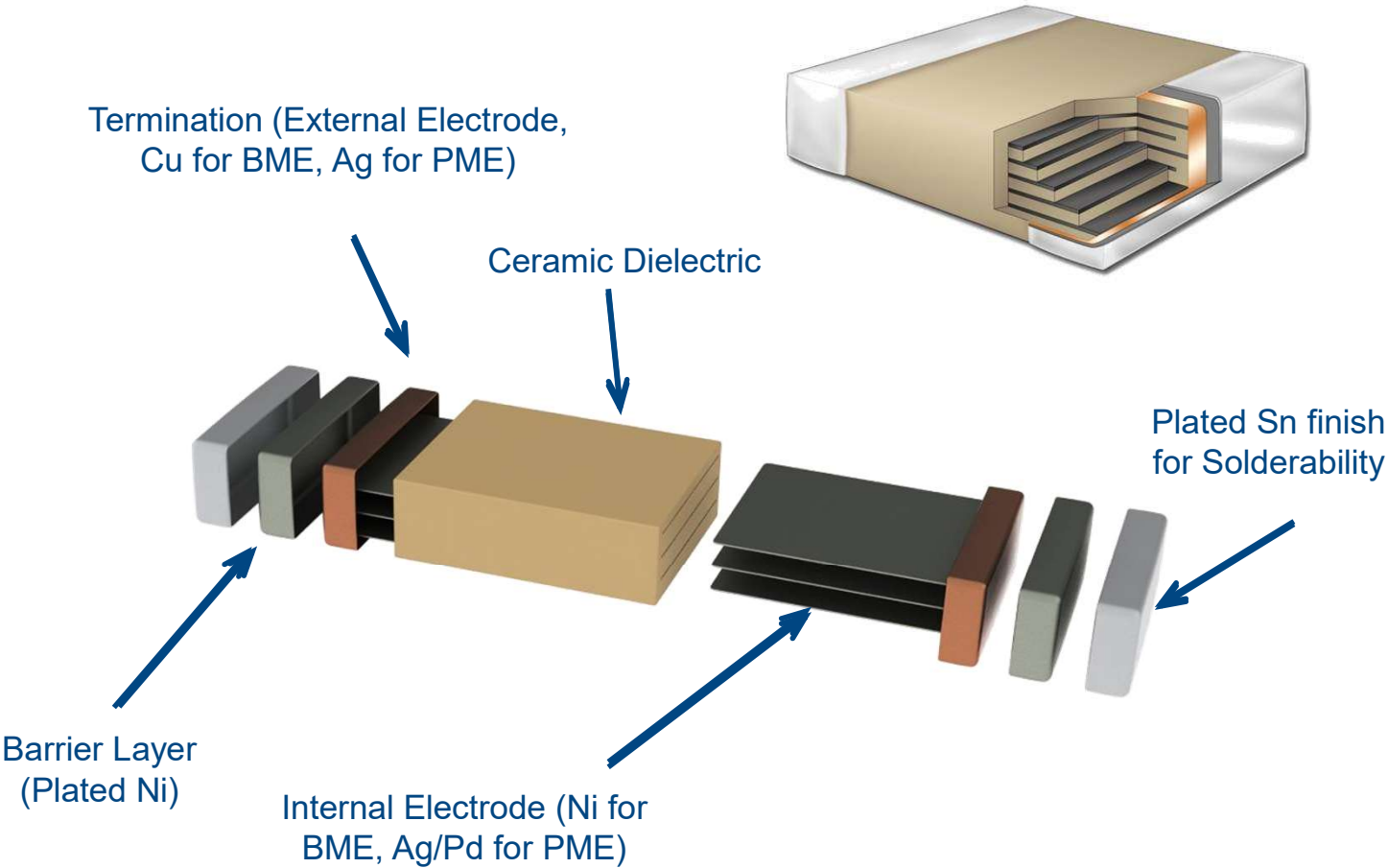
Capacitances in parallel are additive

$$C_T = C_1 + C_2 + C_3 + \dots C_n$$

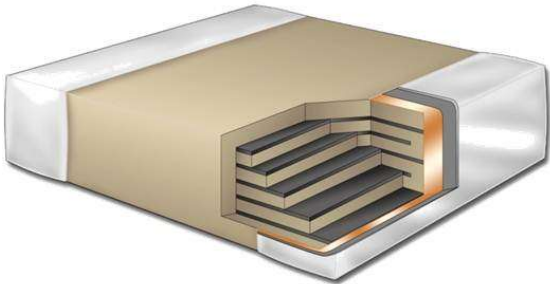
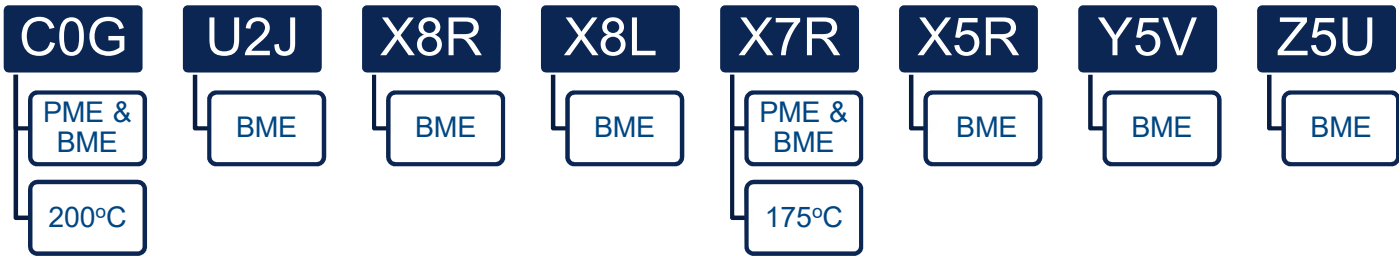
$$C = \frac{\epsilon_0 K A (n-1)}{d}$$

Multilayer Ceramic Capacitor (MLCC)

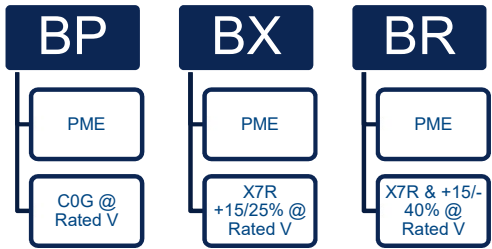
Typical Construction



Commercial & Automotive Grade Dielectric Materials

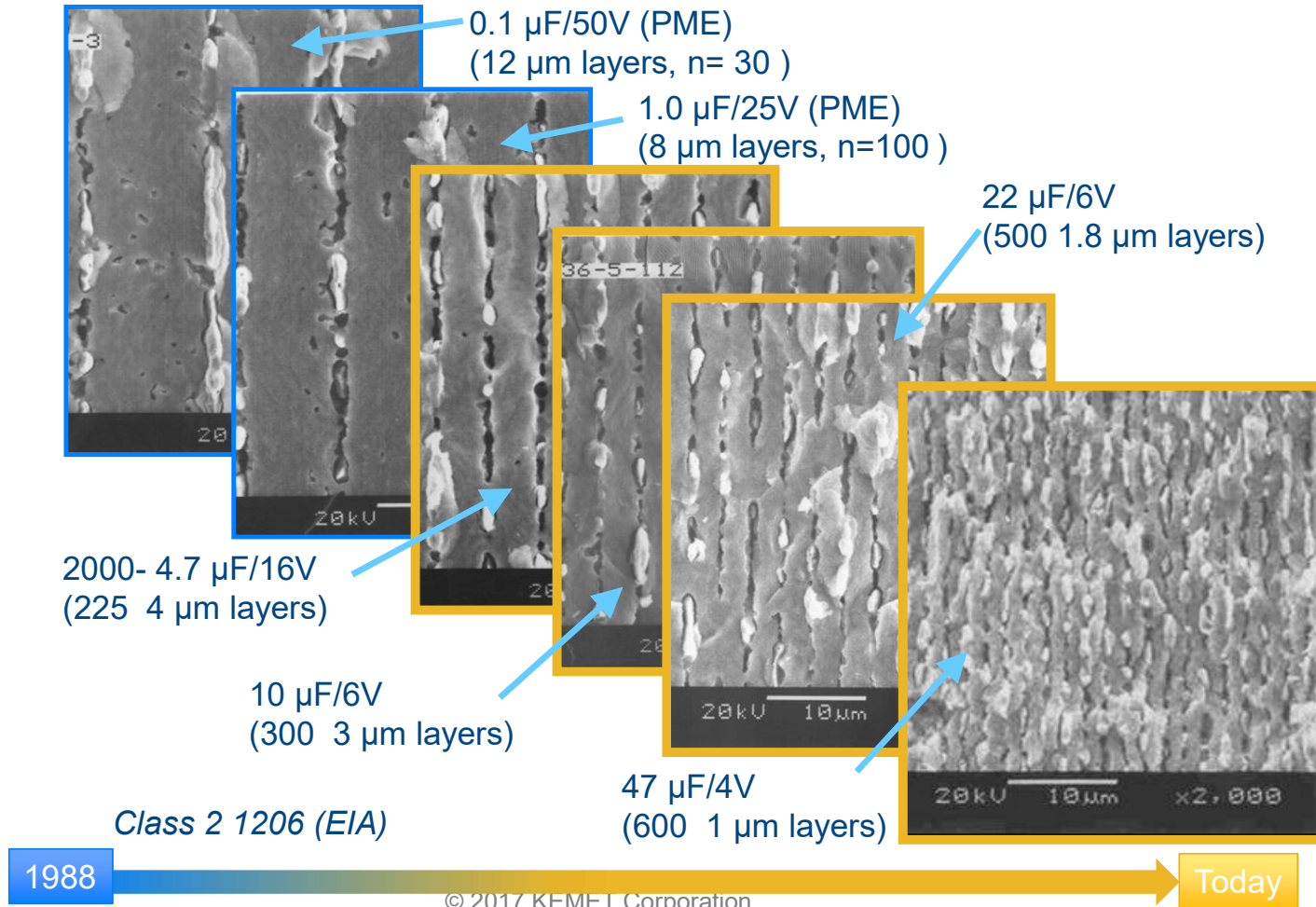


Military & Hi-Rel Dielectric Materials

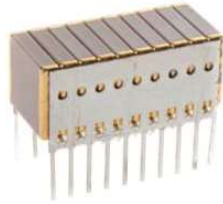


Trend in BME MLCC Technology:

Dielectric Thickness and Layers Count Progression



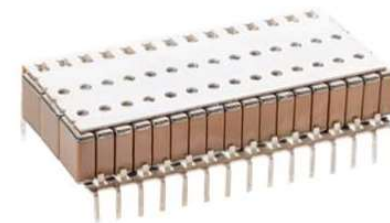
Ceramic Engineered Solutions

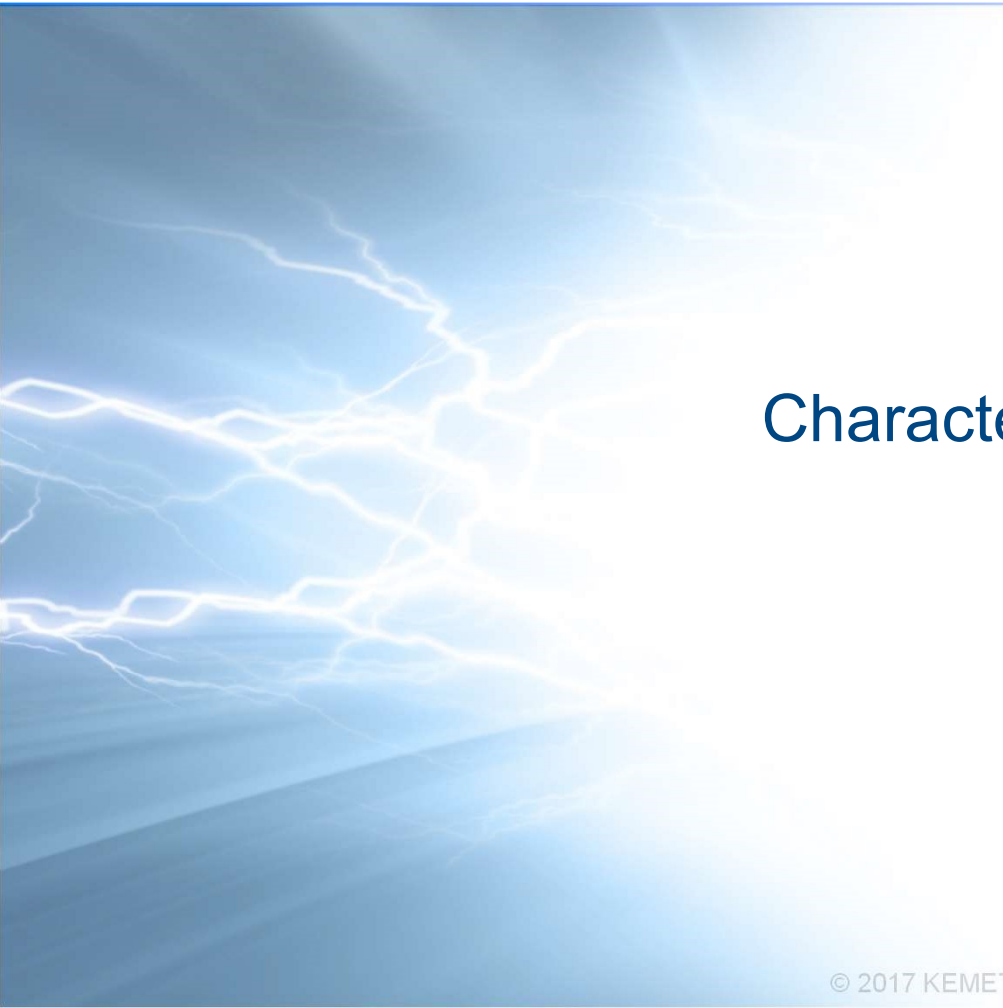


- RoHS and Non-RoHS
- Extensive Dielectric Portfolio
- Bulk Capacitance
- High Voltage
- High Temperature
- SMD & Through-Hole
- Non Standard Sizes and Configurations
- A full range of termination materials and finishes



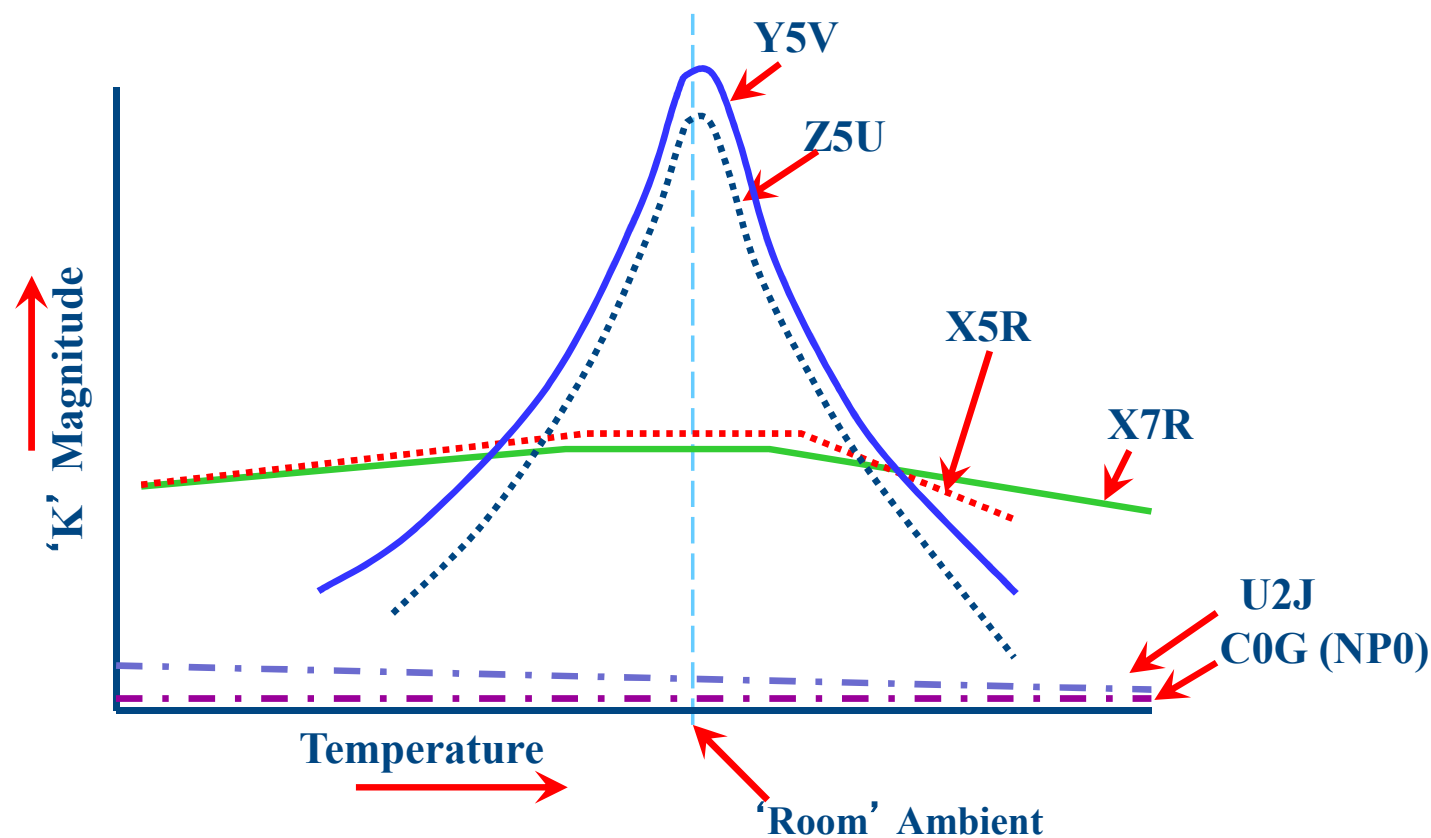
- Arc Prevention
- Flex Mitigation
- ESD
- Noise Reduction
- Pulse Capable
- High Shock & Vibration
- Integrated Technology
- Specialized Testing/ Screening
- Encapsulation





Characteristics

Relative Capacitance vs. Temperature



Dielectric Classification

Class I (Per EIA – 198)

Class I Dielectrics: (Example: C0G)

Alpha Symbol	Significant Figure of Temp Coefficient ppm/°C	Numerical Symbol	Multiplier to significant figure	Alpha Symbol	Tolerance of Temp Coefficient ± ppm/°C
C	0	0	-1	G	30
B	0.3	1	-10	H	60
L	0.8	2	-100	J	120
A	0.9	3	-1000	K	250
M	1.0	4	-10000	L	500
P	1.5	5	+1	M	1000
R	2.2	6	+10	N	2500
S	3.3	7	+100		
T	4.7	8	+1000		
U	7.5	9	+10000		

Temperature Range: -55°C to +125°C
C0G provides highest temperature stability

Dielectric Classification

Class II and III (per EIA-198)

Alpha Symbol	Low Temperature (°C)	Numerical Symbol	High Temperature (°C)	Alpha Symbol	Max cap change over temp. range (%)
Z	+10	2	+45	A	±1.0
Y	-30	4	+65	B	±1.5
X	-55	5	+85	C	±2.2
		6	+105	D	±3.3
		7	+125	E	±4.7
		8	+150	F	±7.5
		9	+200	P	±10
				R	±15
				S	±22
				* L	+15 to - 40
				T	+22 to - 33
				U	+22 to - 56
				V	+22 to - 82

CLASS II

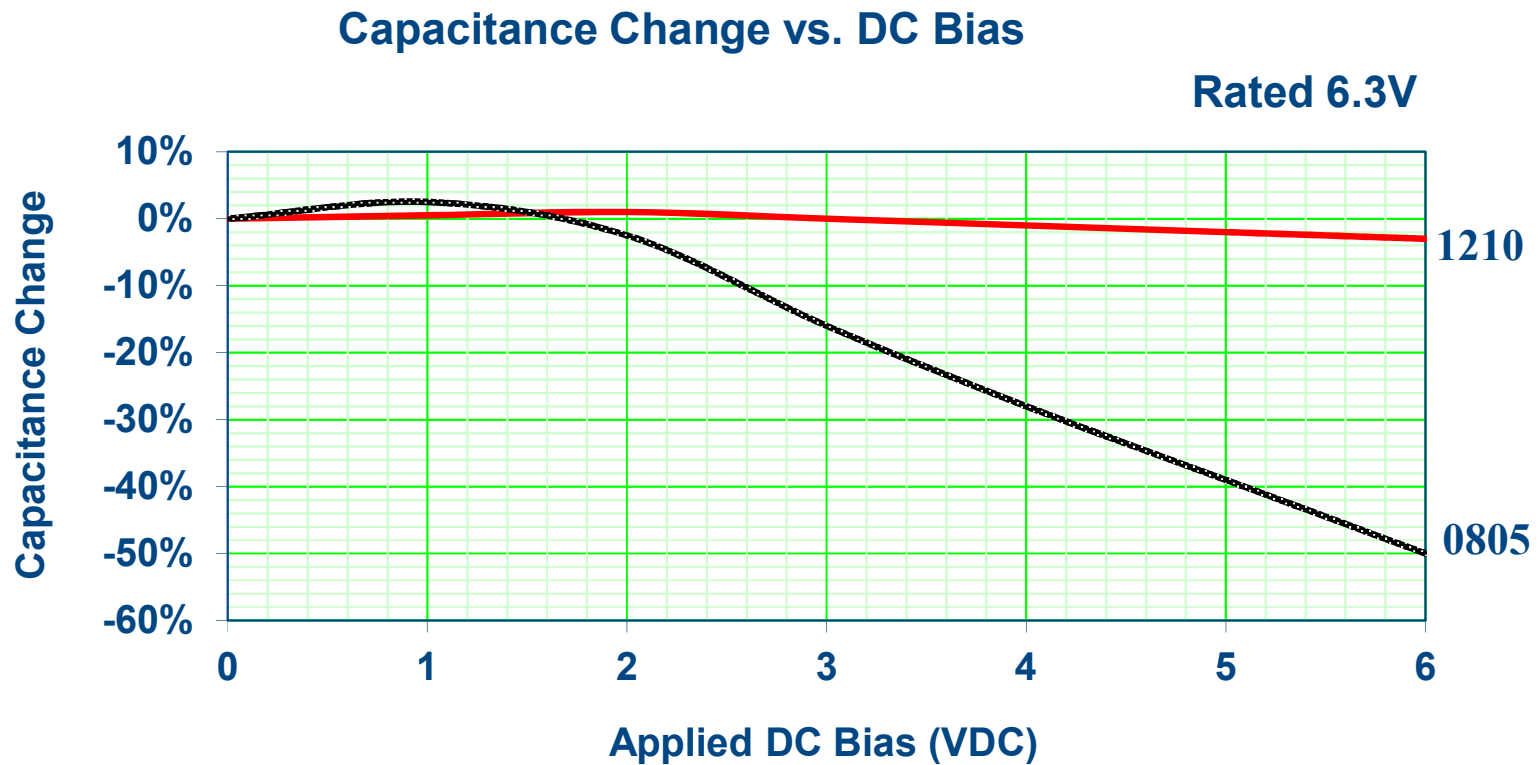
CLASS III

* Industry Classification (Non EIA-198)

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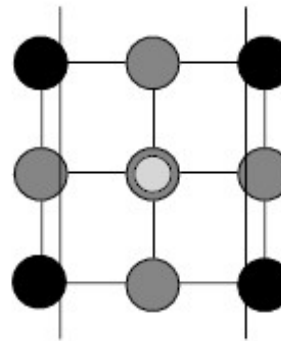
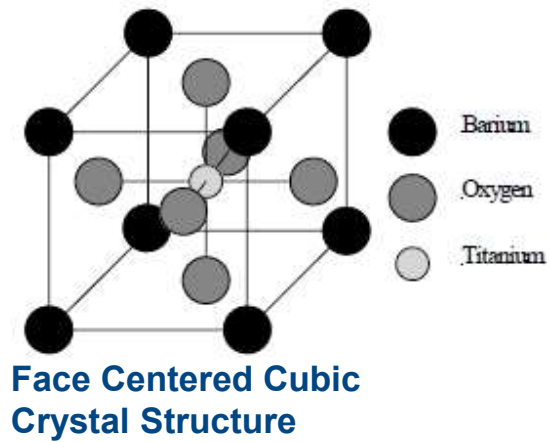
Voltage Coefficient (Class II and III)

1210 vs 0805, X7R, 10uF, 6.3V



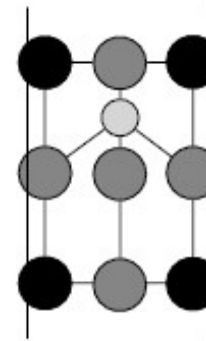
Voltage Coefficient (Class II and III)

DC Bias



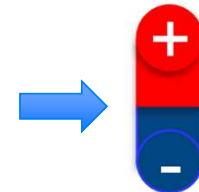
BaTiO₃ above 130°C

- Cubic
- No Dipole

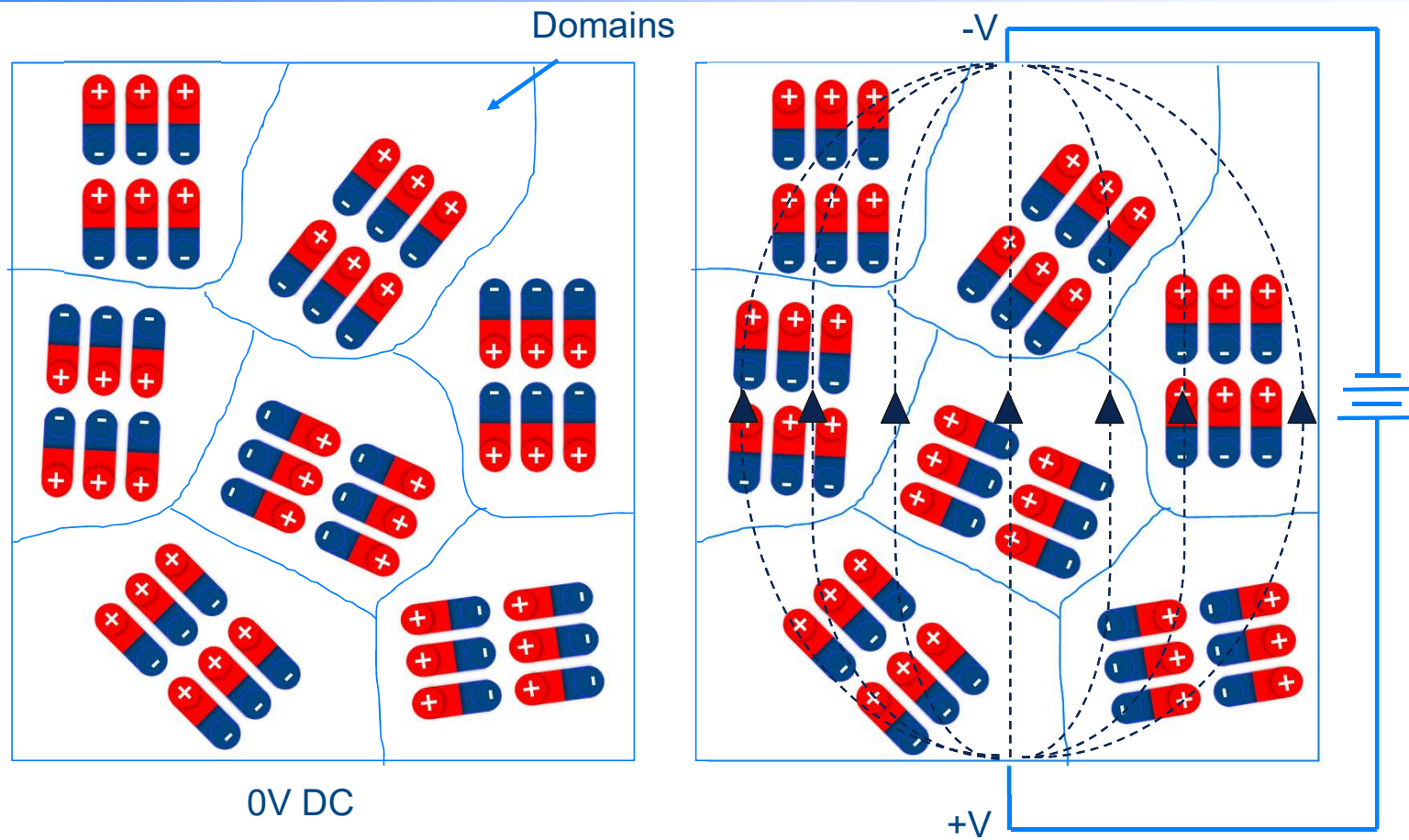


BaTiO₃ below 130°C

- Tetragonal
- Creates Dipole

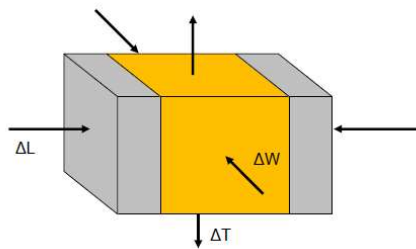
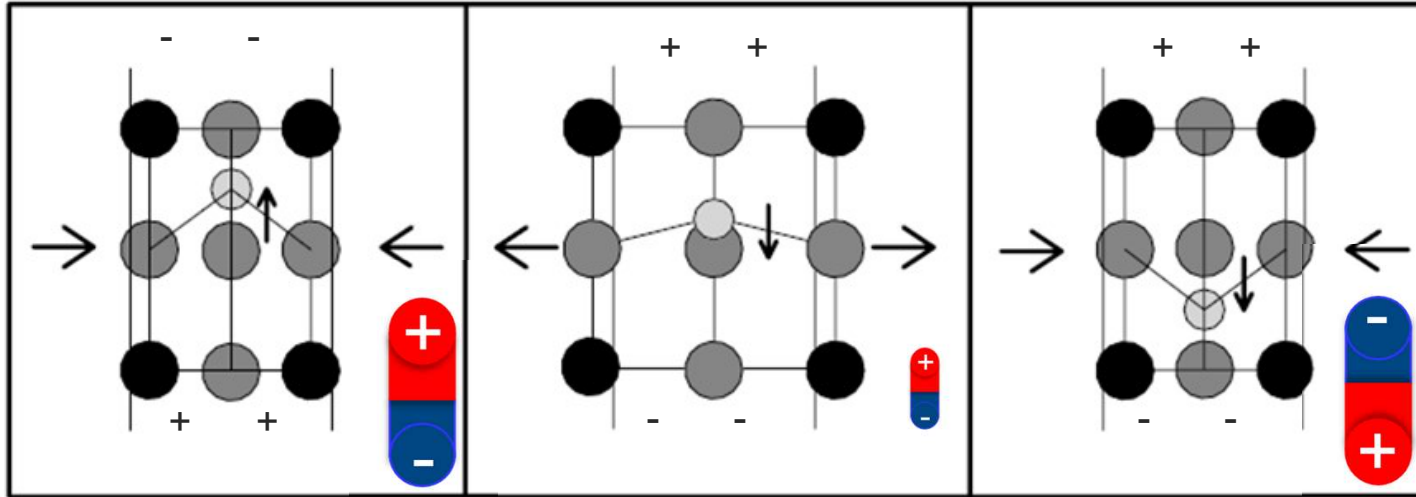


Voltage Coefficient (Class II and III)



Voltage Coefficient (Class II and III)

Piezoelectricity and Electrostriction



Mechanical Distortion

Piezoelectricity and Electrostriction

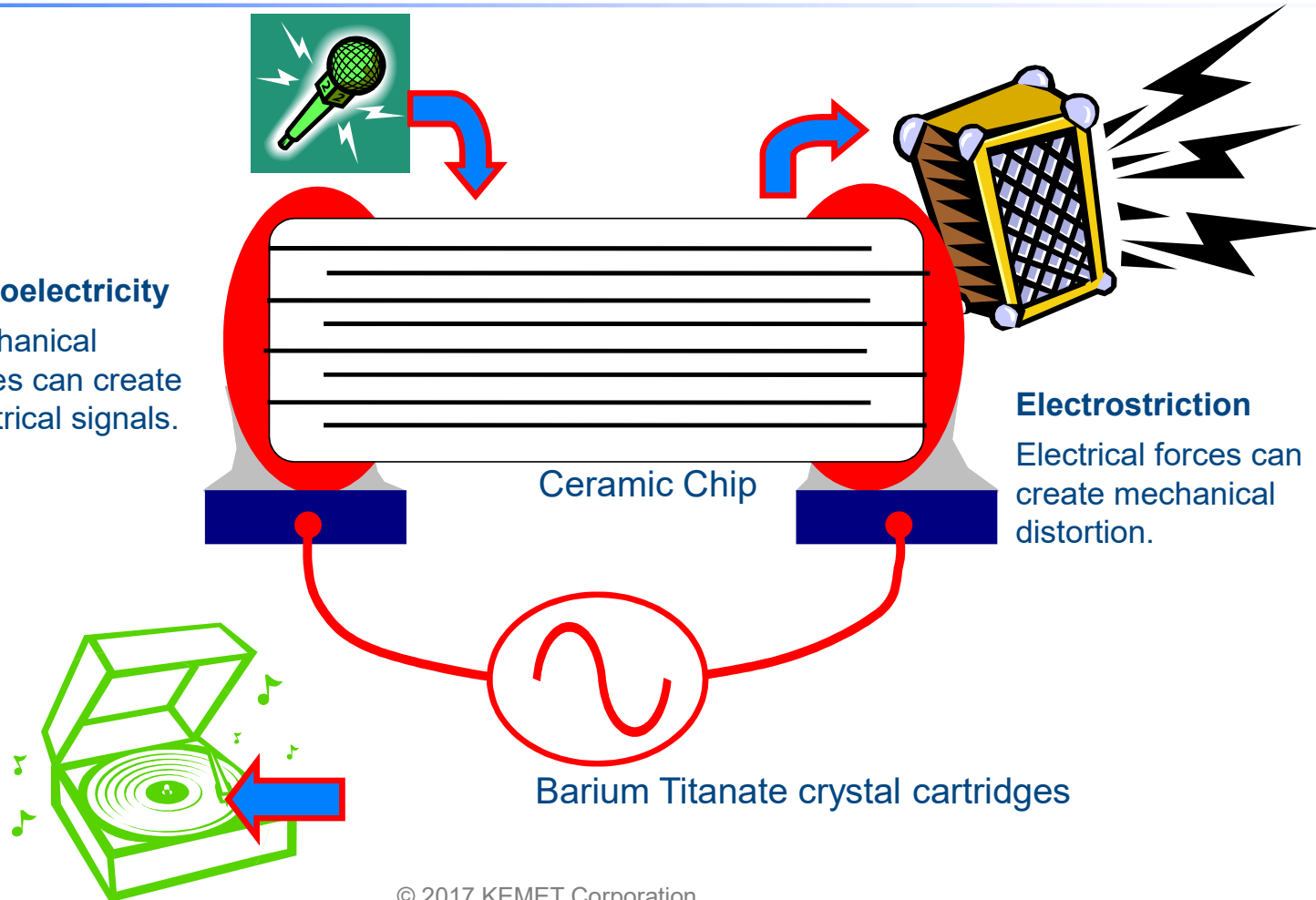
Class II and III Only

Piezoelectricity

Mechanical forces can create electrical signals.

Electrostriction

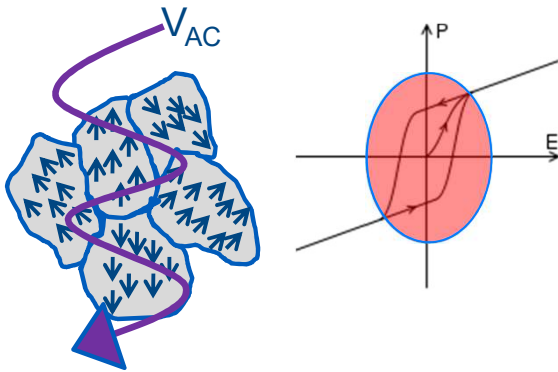
Electrical forces can create mechanical distortion.



AC Coupling and Signal Distortion

X7R vs. C0G

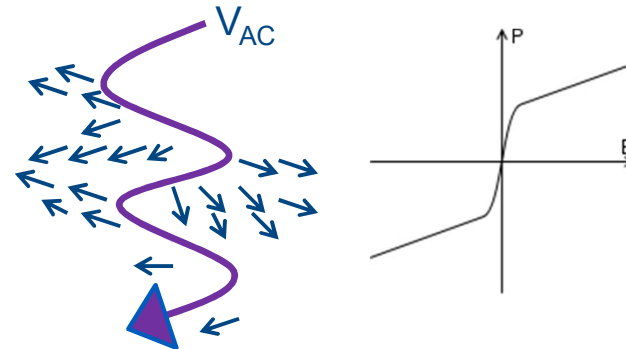
Class 2 BaTiO₃
Ferroelectric



Ferroelectric dipoles in *domains* align with the AC Field

Domain wall heating &
Signal distortion

Class 1 CaZrO₃
Paraelectric



Paraelectric dipoles align with AC field

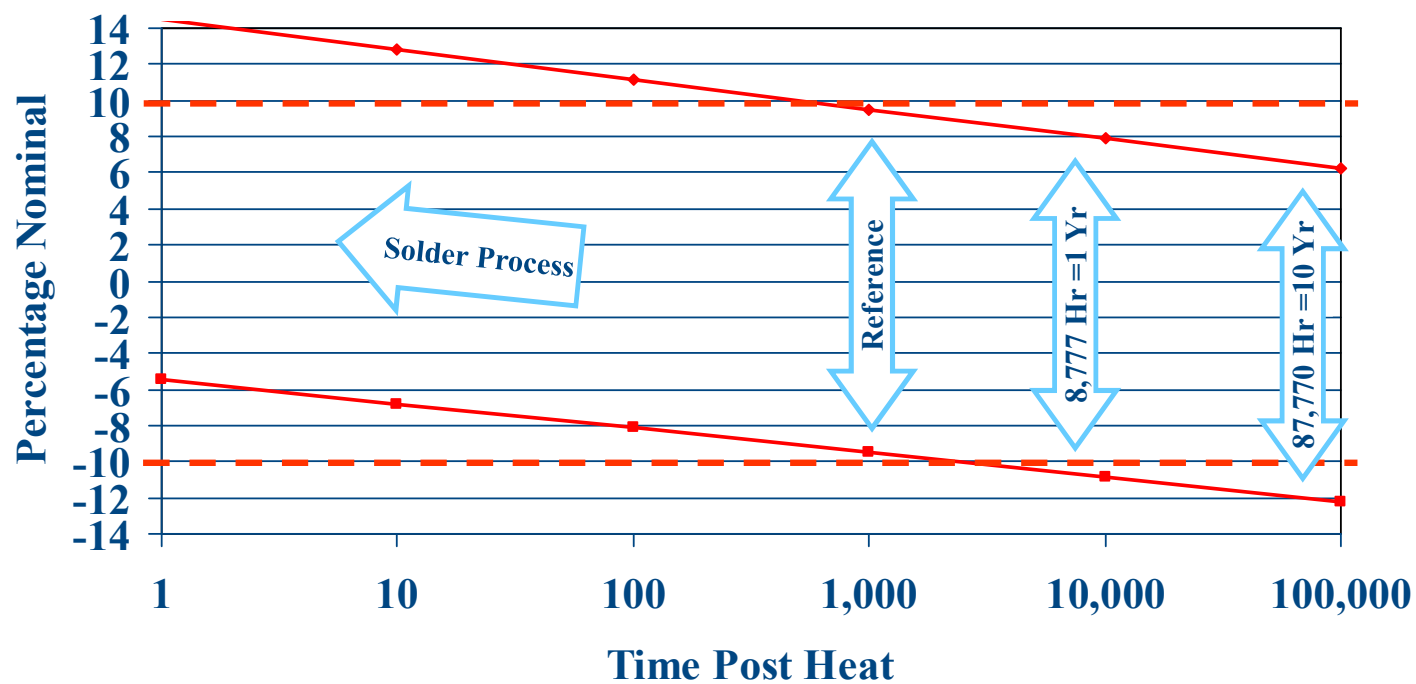
No domains, so
No Domain wall heating &
Reduced signal distortion



Aging of Class 2 and Class 3 Capacitors

X7R Aging Rate

1.5% per Decade Hour (Limit)



<https://ec.kemet.com/design-tools/aging-calculator-for-ceramics>



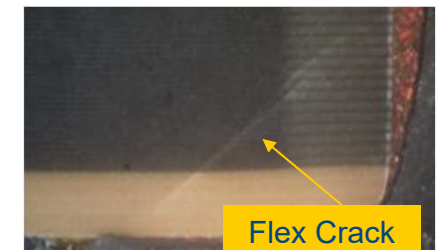
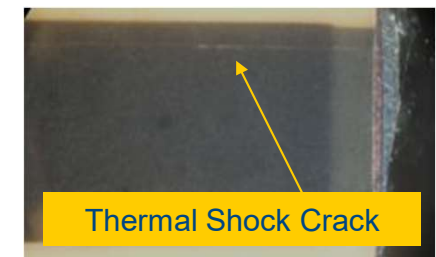
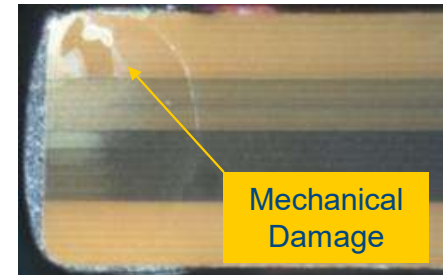
Common Failure Modes

Typical Crack Signatures

MLCC Cross-Sections

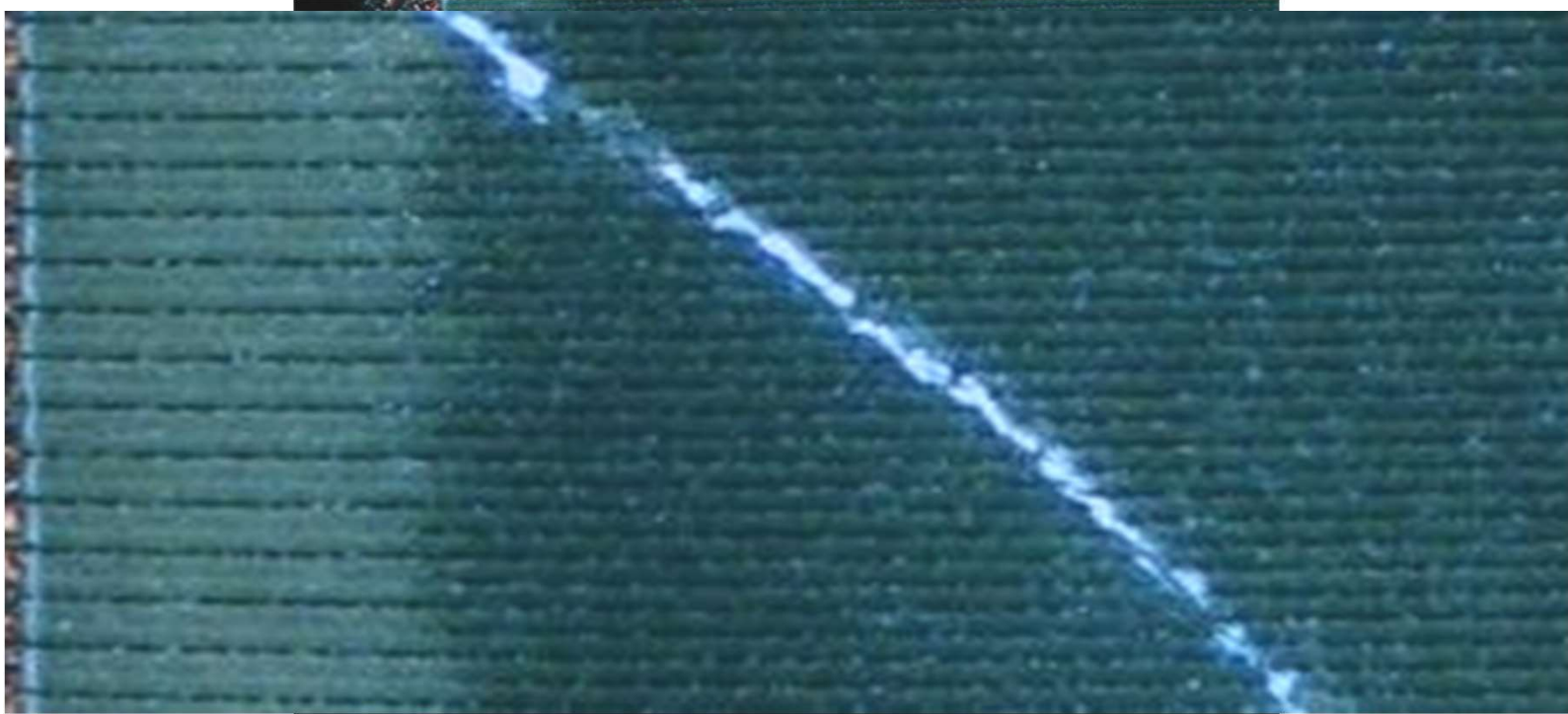
The major sources MLCC of cracks are:

- Mechanical damage (impact)
 - Aggressive pick and place
 - Physical mishandling
- Thermal shock (parallel plate crack)
 - Extreme temperature cycling
 - Hand soldering
 - *Do not touch electrodes while hand soldering!*
- Flex or Bend stress
 - Occurs after mounted to board
 - Common for larger chips (>0805)



Failure is **not always** immediate!
Failure mode is **not always** deterministic!

Flex Cracks



<https://ec.kemet.com/q-and-a/what-is-failure-mode-for-ceramic-capacitors>

<https://ec.kemet.com/knowledge/flexible-termination-reliability-in-harsh-environments>

Flex Mitigation Technology

Select the Right Level of Protection for Your Application



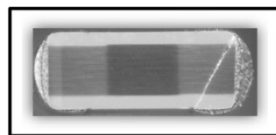
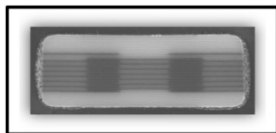
Level 0: NO Crack Protection

Standard MLCC

Target Applications:
Non-Critical

Fail-Short Condition

Up to 2mm flex bend capability



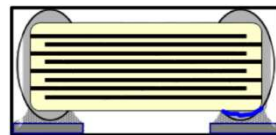
Level I: Basic Level of Crack Protection

Floating Electrode or Open-Mode

Target Applications:
Semi - Critical

Fail-Open Condition

Up to 2mm flex bend capability



Level II: Intermediate Level of Crack Protection

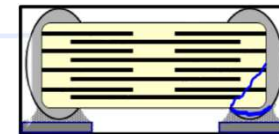
Flexible Termination

Target Applications:
Critical

Flexible termination provides for a high level of protection from thermal stress cracks, pick-and-place damage, and board flex stress

Fail-short Condition

Up to 5mm flex bend capability.



Level III: High Level of Crack Protection

Floating Electrode plus Flexible Termination

Target Applications:
Safety Critical

Combines cascading electrode design with tear-away, termination technology. Provides for a high level of protection from thermal stress cracks, pick-and-place damage, and board flex stress

Fail-Open Condition

Up to 5mm flex bend capability.



Capacitors for RF Applications

RF Capacitor Basics

Some Key Parameters

ESR

Effective Series Resistance

- The resistance of the capacitor which includes resistance due to the dielectric as well as electrodes.

Q

Quality Factor

- Quantifies the amount of energy stored versus how much is dissipated as heat. It represents the efficiency of the capacitors. Higher Q's are needed for RF capacitors to limit power dissipation.

SRF

Series Resonant Frequency

- Shows where the total impedance is no longer capacitive and begins an upward trend (becomes inductive). Higher SRF = better RF capacitor, since some applications require the designer to stay well below the SRF.

TCC

Temperature Coefficient of Capacitance

- Determines how much the capacitance values will shift at different temperatures. RF capacitors need to be very stable over a broad temperature range.

$$Q = \frac{X_c}{ESR} = \frac{1}{DF}$$

$$SRF = \frac{1}{2 * \pi * \sqrt{C * L}}$$

C0G → ppm / °C level
X7R → % level

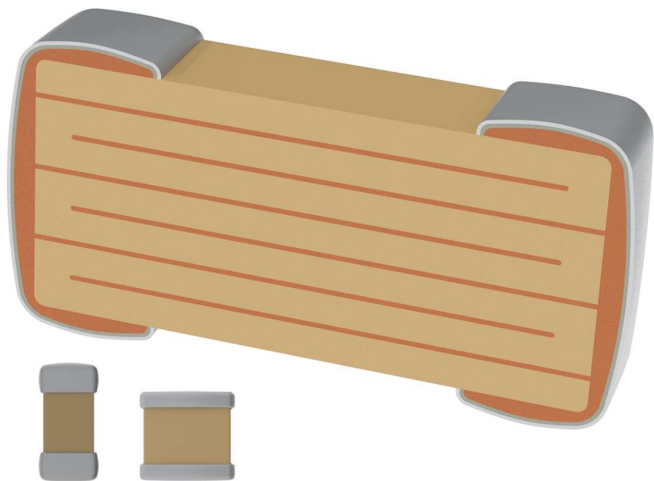
What is an RF Capacitor?

An RF capacitor is a capacitor whose “characteristics” are favorable at RF frequencies.

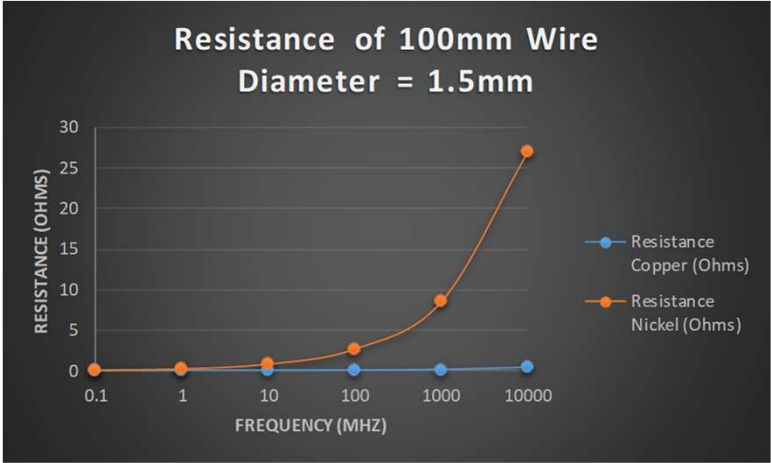
Characteristic	RF Capacitor Requirements
ESR (Effective Series Resistance)	RF Capacitors are designed to have the lowest possible ESR. This allows for minimal power loss at RF frequencies.
Q (Quality Factor)	RF Capacitors are designed to have a high Q.
SRF (Series Resonant Frequencies)	RF Capacitors are designed to have high SRF allowing for a higher operating frequency range.
TCC (Temperature Coefficient of Capacitance)	The dielectric is chosen to have a minimal capacitance shift across its entire operating temperature range.

So, for RF capacitors, materials are chosen and the design is optimized so that the capacitors' characteristics are well suited at the higher frequencies.

RF Capacitor Construction



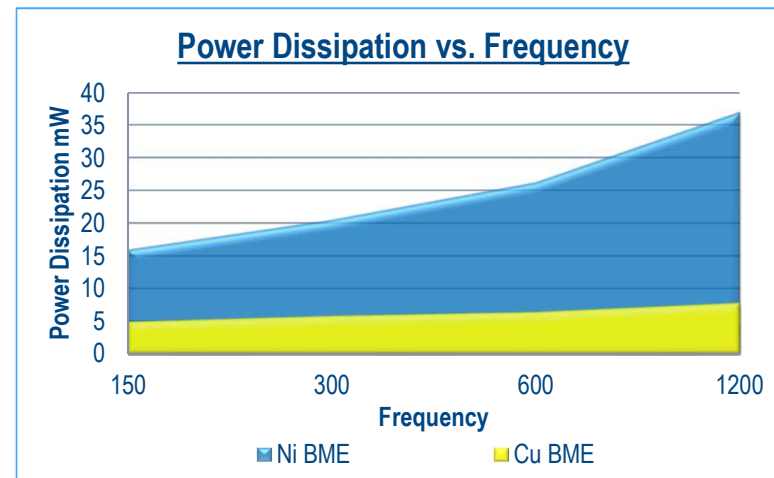
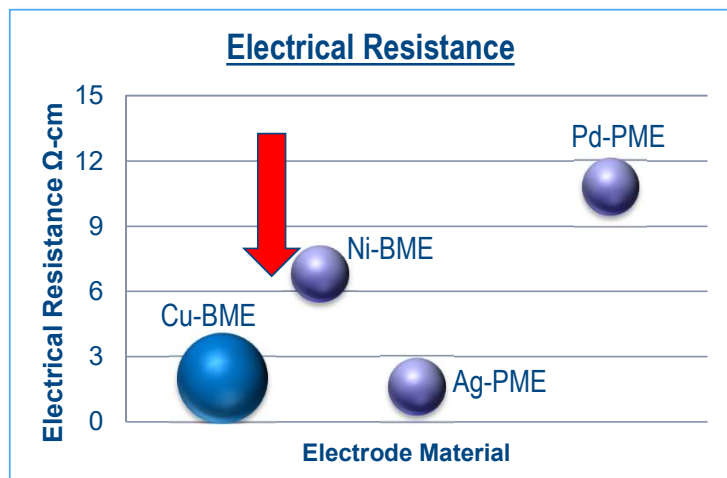
Long/Narrow vs. Short/Wide



Design	Characteristic
Dielectric	Low-loss dielectrics are chosen to reduce ESR. Typically, these are C0G dielectrics which also provide temperature stability (TCC) performance.
Electrodes	Electrode materials are chosen to provide the lowest ESR and ESL over a broad frequency range. This means we stay away from ferrous materials such as nickel.
Construction / Physical Geometry	Physical geometry plays an important role in resistance and inductance of the capacitor. Long and narrow capacitors will have a higher ESR and ESL than a short and wide capacitor.

RF Capacitors

Why Copper BME?



Copper BME = Lower ESR = Better power dissipation = **Ideal for High Frequency applications**

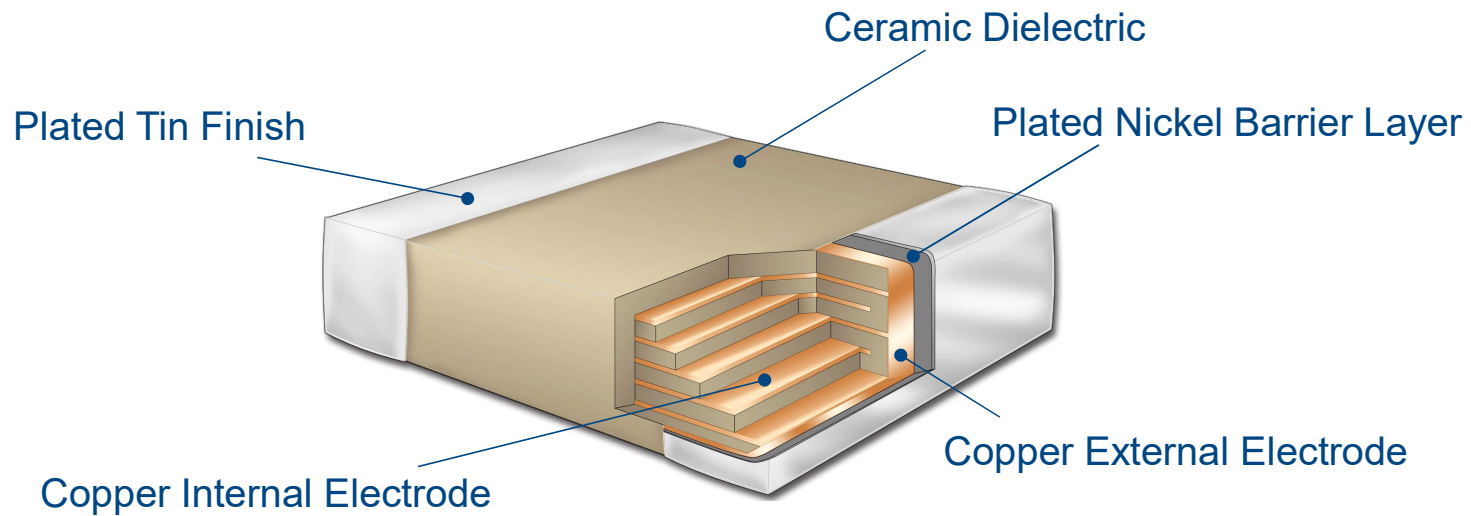
Application	Power and Frequency Capabilities	
Base Station / Power Amp	C0603-C2225 <10W Power <100MHz Frequency	CBR06, CBR08 >10W Power >100MHz Frequency
Mobile Phone	C0201, C0402 <1W Power <100MHz Frequency	CBR02, CBR04 >10W Power >100MHz Frequency
	COMMERCIAL	RF & MICROWAVE

KEMET CBR RF Capacitors

Construction

Electronic Components
KEMET
CHARGED.®

 **HiQ-CBR**
RF & MICROWAVE



Base Metal, **Copper Electrodes**

Thank You!